

READING PASSAGE 3

Nuclear Energy and Waste Management

A In recent years, nuclear energy and waste management has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D Case studies from diverse geographical and cultural settings provide compelling illustrations of these dynamics in action. In one particularly well-documented example, researchers tracked developments over a fifteen-year period in a setting that offered a natural experiment in the effects of policy change. The introduction of a new regulatory framework produced a cascade of adaptations that rippled through the entire system, generating outcomes that both confirmed theoretical predictions and revealed previously unrecognised complexities.

E Despite the substantial progress that has been made, significant challenges and controversies remain. Critics have raised important methodological concerns regarding the generalisability of findings derived from studies conducted in specific contexts. The predominance of research carried out in Western, educated, industrialised, rich, and democratic societies raises questions about the applicability of current theories to populations with different cultural, economic, and political circumstances. Furthermore, the reliance on cross-sectional designs in much of the existing literature limits the capacity to draw strong causal inferences.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Choose the correct letter, A, B, C or D.

- 1 According to the text, what role do structural factors play?
- A. inconsistent results across cultures
 - B. a consistent pattern across diverse contexts
 - C. no measurable effects
 - D. effects limited to Western countries

Question 2

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 2 The author believes that all research questions in this field have been answered.

Questions 3–6

Choose the correct letter, A, B, C or D.

- 3 According to the passage, what is one of the key factors identified by researchers?
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations
- 4 According to the text, what role do structural factors play?
- A. minimal impact
 - B. creating underlying conditions for action
 - C. determining individual choices directly
 - D. irrelevant to outcomes
- 5 Which of the following best describes the author's view on future developments?
- A. minimal impact
 - B. creating underlying conditions for action
 - C. determining individual choices directly
 - D. irrelevant to outcomes
- 6 The passage indicates that critics have raised concerns about:
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Questions 7–8

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

7 Policy-makers have fully implemented all research recommendations.

8 Meta-analyses have confirmed the reliability of the core findings in this field.

Question 9

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

9 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 10

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

10 The author believes that all research questions in this field have been answered.

Questions 11–13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

12 The landmark study analysed data spanning a period of _____ years.

13 Case studies have documented the effects of _____ change on the system.